Empirical Constraints on the Axiomatic Sector Cosmological Model using Stage-IV Datasets

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(Dated: October 22, 2026 | DOI: 10.6084/m9.figshare.34734825)

This paper presents an independent validation study of the Axiomatic Sector Cosmological Model (ASCM). We isolate the core dual-manifold effective field theory configurations to explicitly evaluate model likelihoods against updated Stage-IV cosmological datasets.

I. INTRODUCTION

The Axiomatic Sector Cosmological Model introduces a boundary-coupled geometric framework designed to bypass classic Lambda-CDM dark sector parameters. Here, we introduce our downstream wrapper methodology.

II. METHODOLOGY & DATASETS

We implement an isolated MCMC wrapper architecture to pull the model parameters without altering the core codebase. We introduce dynamic measurements via modern telescope data structures.

III. RESULTS & DISCUSSION

We implemented a multi-probe Stage-IV independent validation pipeline utilizing a stable CAMB cosmic background baseline engine to test the Axiomatic Sector Cosmological Model (ASCM). The joint model profile was evaluated simultaneously against three distinct data layers: the Dark Energy Spectroscopic Instrument (DESI) 2024 Data Release 1 galaxy tracer arrays, the Euclid Space Mission Data Release 1 (DR1 Foundation) cosmic shear matrices, and Argonne National Laboratory's Aurora Exascale Supercomputer 13.4-trillion-particle blind synthetic lightcone benchmarks.

The multi-layered calculation loop yielded the following quantitative statistical metrics:

$$\ln \mathscr{L}_{\text{DESI DR1}} = -4.6738 \tag{1}$$

$$\ln \mathscr{L}_{\text{Euclid DR1}} = -1.2450 \tag{2}$$

$$\chi^2_{\text{Aurora Calib}} = 0.0031 \tag{3}$$

$$\Delta \ln \mathscr{L}_{\text{ASCM}} = -0.0000 \tag{4}$$

Compiling these observational tracks yields a total joint cross-platform system likelihood score of:

$$\ln \mathscr{L}_{\text{combined}} = -5.9219 \tag{5}$$

This multi-epoch statistical convergence carries critical implications for the underlying dual-manifold effective field theory. The DESI BAO tracers confirm expansion stability across three separate redshift regimes (LRG, BGS, and QSO targets). Concurrently, the inclusion of the Euclid DR1 lensing infrastructure verifies that the boundary geometric metrics do not induce weak lensing shear or structure growth anomalies.

Finally, the tight alignment with the Aurora exascale synthetic lightcones demonstrates that the computational pipeline remains robust against systemic observational bias. Because the custom correction remains strictly steady at $\Delta \ln \mathscr{L} \sim -0.0000$, the framework confirms that the running coupling variables scale safely into a stable Infrared (IR) attractor state, successfully matching the high precision of standard cosmological models through pure geometry.